Question	Answer	Marks
5(a)	work done per unit charge	B1
	work done (on charge) in moving positive charge from infinity (to the point)	B1
5(b)(i)	electric field strength inversely proportional to distance^2	C1
	$E = 3^2 \times 2.0 \times 10^5$ $= 1.8 \times 10^6\ \text{N C}^{-1}$	A1
5(b)(ii)	$V = Q / 4\pi\varepsilon_0 r$ **and** $E = Q / 4\pi\varepsilon_0 r^2$ (so $E = V / r$)	B1
	$r = (9.0 \times 10^4) / (1.8 \times 10^6) = 0.050\ \text{m} = 5.0\ \text{cm}$	A1
5(b)(iii)	$Q = 4\pi\varepsilon_0 Vr$ $= 4\pi \times 8.85 \times 10^{-12} \times 9.0 \times 10^4 \times 0.050$	C1
	$= 5.0 \times 10^{-7}\ \text{C}$	A1
5(b)(iv)	$C = Q / V$	C1
	$= (5.0 \times 10^{-7}) / (9.0 \times 10^4)$ $= 5.6 \times 10^{-12}\ \text{F}$	A1

Question	Answer	Marks